

Jayamini Dissanayake

Undergraduate – Computer Systems Engineering
Sri Lanka Institute of Information Technology (SLIIT)
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PROFILE

Undergraduate in Computer Systems Engineering at Sri Lanka Institute of Information Technology with hands-on experience in software development, embedded systems, digital logic design, and wireless system simulations. Skilled in full-stack web development and control systems with a strong passion for solving real-world engineering problems. Seeking an internship or entry-level opportunity to apply technical knowledge, enhance practical skills, and contribute to innovative technology solutions.

EDUCATION

BSc (Hons) in Computer Systems Engineering

Sri Lanka Institute of Information Technology (SLIIT)

2024 – 2027

G.C.E. Advanced Level (A/L)

Physical Science Stream

2022

TECHNICAL SKILLS

Programming Languages - Python, C, C++, Java, JavaScript, SQL

Embedded Systems - Raspberry Pi, Arduino, AVR Concepts, Sensor Integration, Microcontrollers

Control Systems - PID Control, Feedback Systems, Real-Time Control Systems

Web Development - Spring Boot, React, REST APIs, HTML, CSS, MySQL

Tools & Platforms - Linux, GitHub, VS Code, Eclipse, Google Cloud Platform, EasyEDA

Concepts - System Integration, Hardware–Software Interfacing, Automation, Embedded Systems Design, OOP, PCB Design, Debugging, Troubleshooting, System Testing

PROJECT & TECHNICAL EXPERIENCE

PID-Controlled DC Motor Stabilization System

April 2026

- Designed and implemented a PID-controlled DC motor speed stabilization system using Raspberry Pi and Python in a Linux environment.
- Developed closed-loop motor control with encoder-based real-time speed feedback and sensor processing.
- Tuned PID parameters (K_p , K_i , K_d) to achieve optimized system stability, low overshoot, and minimal steady-state error.
- Designed and integrated a custom PCB using EasyEDA for compact and reliable hardware implementation.
- Performed system modeling, gain tuning, disturbance testing, and performance evaluation for control system optimization.

Technologies: Python, Raspberry Pi, PID Control, EasyEDA

Line Following Arduino Robot Car

April 2026

- Designed and developed an autonomous line-following robot car using Arduino and embedded system principles.
- Implemented sensor-based path detection and real-time motor control for autonomous navigation.

- Developed embedded control logic using C/C++ and integrated IR sensors for accurate line tracking.
- Designed and tested hardware interfaces, motor driver control, and obstacle handling mechanisms.
- Applied embedded systems engineering concepts including sensor integration, control logic, and real-time system testing.

Technologies: Arduino, IR Sensors, Embedded C/C++

CradleBot – Smart Automated Baby Cradle System

April 2024

- Designed and developed an automated smart cradle system capable of detecting baby cries and initiating automatic rocking motion.
- Integrated sound and moisture sensors for infant monitoring and responsive system control.
- Implemented servo motor-based cradle movement and MP3 audio playback for soothing lullaby functionality.
- Developed embedded control logic using Arduino for real-time sensor processing and automation.
- Applied embedded systems, sensor interfacing, and automation principles to improve infant comfort and parental convenience.

Technologies: Arduino, Servo Motors, Sensors

AM Broadcast Receiver

October 2025

- Designed, built, and tested the audio output circuit of a basic AM broadcast receiver using discrete electronic components.
- Implemented multi-stage analog circuits including band-pass filtering and BJT-based signal amplification for audio recovery.
- Designed a common-emitter BJT amplifier with stable biasing to amplify weak detector signals for audio output.
- Performed circuit analysis, component selection, waveform testing, and signal observation using oscilloscope measurements.
- Constructed and tested the complete hardware circuit on a dot board with reliable wiring and circuit integration

Technologies: Analog Electronics, BJT Amplifiers, Filters, Oscilloscope Testing, Circuit Design, Dot Board Implementation

CERTIFICATIONS

Network Support & Security – Cisco

Web Design for Beginners – University of Moratuwa

Python Programming – University of Moratuwa

AI/ML Engineer (Stage 1 & 2) – SLIIT

Software Architecture – Online

NMAP: Network Scanning & Recon (CNMP+) – OCSALY

SOFT SKILLS

Time Management

Communication Skills

Adaptability and Quick Learning

Teamwork and Collaboration

Problem-Solving and Analytical Thinking

Long term planning

References

Dr. Kapila D. Dissanayaka

Senior Lecturer (HG), Faculty of Computing

Sri Lanka Institute of Information Technology (SLIIT)

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Ms. Dinithi S. Pandithage

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